Network Science
PHYS/NETS 5116, Fall 2026

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Assignment 1, due by Friday, Oct. 9th, 11:59pm EST.

Please submit your assignment via Canvas. Your submission should include (1) your solutions to these questions and (2) the Python code used to answer Questions 5 and 6 (preferably as a .ipynb notebook file). Please embed your visualization for Question #3 in the same document as your other solutions. Please name your work as Lastname.pdf, *e.g.* Barabasi.pdf (or Barabasi.ipynb). On Wednesday, October 14th, we will have an in-class assessment based on these questions.

1. *Your Interest.*

Tell us about the network you are personally most interested in. Address the following questions:

(a) What are its nodes and links?

(b) How large is it?

(c) Can it be mapped out?

(d) Why do you care about it?

2. *Matrix Formalism.*

Let $\mathbf{A}$ be the $N \times N$ adjacency matrix of an undirected unweighted network, without self-loops. Let $\mathbf{1}$ be a column vector of N elements, all equal to 1. In other words, $\mathbf{1} = (1, 1, \ldots, 1)^T$, where T indicates the transpose operation. Use matrix formalism (multiplicative constants, multiplication row by column, matrix operations like transpose and trace, etc., but avoid the sum symbol $\sum$) to write expressions for:

(a) The vector $\mathbf{k}$ whose elements are the degrees k_i of all nodes $i = 1, 2, \ldots, N$.

(b) The total number of links, L, in the network.

(c) The number of triangles, n_Δ, present in the network, where a triangle means three nodes, each connected by links to the other two. (Hint: you can use the trace of a matrix.)

(d) The vector k_{nn} whose element i is the sum of the degrees of node i's neighbors.

3. *Visualization.*

Download the `netscience` dataset at this URL: `https://tinyurl.com/bd6cwsd4`. As described in the `.txt` file, this is a coauthorship network of scientists working on

network theory and experiments, compiled by Mark Newman in 2006. Visualize the largest component of the network (Gephi may be the easiest choice of software, but you are free to use your favorite), considering the following:

- Degree or other centrality measures
- Edge properties
- Layout

Make good use of color, size, and layout to create a clear, informative visualization. Describe your approach and comment on your observations (about 5-10 sentences).

4. *Erdős-Rényi Networks.*

Consider an Erdős-Rényi network with $N = 6,000$ nodes, connected to each other with probability $p = 10^{-4}$.

(a) What is the expected number of links, $\langle L \rangle$?

(b) In which regime is the network?

(c) Calculate the probability p_c such that the network would be at the critical point.

(d) Given the linking probability $p = 10^{-4}$, calculate (or compute) the number of nodes N_{cr} that a given network would need to have in order for all of its nodes to be in a single component. If necessary, round to the nearest integer.

(e) For the network in (d), calculate the average degree $\langle k_{cr} \rangle$ and the average distance between two randomly chosen nodes $\langle d \rangle$.

(f) Calculate the degree distribution p_k of this network (approximate with a Poisson degree distribution). (Use the original network, not the one used in (d) and (e)).

(g) Using the degree distribution in (f), calculate the probability of a node i having degree $k_i = 4$.

5. *Randomizing a network.*

Download the `airport-network.gml` dataset at this URL: `https://tinyurl.com/4rfrx4p9`. Using the `networkx` library in Python, load the dataset with the function `nx.read_gml()` and perform the following analyses.

(a) Make a log-log plot of the degree distribution and compute the global clustering coefficient of this network.

(b) Randomize the network using a "full randomization" (see Fig. 4.17) and recompute (and plot) the degree distribution and report the new global clustering coefficient.

(c) Randomize the network using a "degree-preserving randomization" (see Fig. 4.17) and recompute (and plot) the degree distribution and report the new global clustering coefficient.

(d) Compare and contrast the degree distribution and global clustering coefficient across the different randomizations and original network.

6. *Watts & Strogatz, 1998.*

The "Watts-Strogatz" model[1] is a simple random network model that produces graphs that have 1) the small world property and 2) high clustering (see Box 3.9). Reproduce the canonical figure[2] from the original 1998 article (also visualized in subplot **d** of Fig. 3.14 in Box 3.9), and briefly comment on the key insights from the figure.

[1] Watts, D. & Strogatz, S. Collective dynamics of 'small-world' networks. *Nature* **393**, 440–442 (1998). `https://doi.org/10.1038/30918`.

[2] `https://www.nature.com/articles/30918/figures/2`